

Question Paper

Exam Date & Time: 29-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025

Digital Forensics [IT_3129]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Describe the major challenges investigators face in computer crime investigations. Illustrate with one real-life case study. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Remember"
- B) During an identity theft investigation, a suspect's laptop and smartphone are seized. Explain the main computer forensics techniques that can be applied to uncover evidence of the crime. (5)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- 2) Describe the key roles and responsibilities of a skilled Incident Response (IR) team in managing and mitigating security incidents. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Understand"
- B) A company reports unauthorized access to its confidential files. As a forensic investigator, illustrate the key steps involved in the data acquisition process to ensure digital evidence is accurately captured and preserved for analysis. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) While investigating a system intrusion, an analyst must examine a suspect's DOS-based storage media. How forensic tools and commands in DOS can be used to retrieve hidden or deleted files. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Analyze how Write Blockers (hardware tools) help maintain the integrity of evidence during forensic data acquisition. Provide an example of their use in real-world investigations. (5)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) A forensic examiner needs to verify the authenticity of acquired evidence. Analyze how hash algorithms like MD5 or SHA-1 ensure reliable forensic data validation and maintain evidence integrity. (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Analysis"
- B) Define network forensics and describe how it helps in identifying and tracing malicious network activities. (5)

Course Outcome: "CO4" / Cognitive Level: "Understand"

5) How ethical hacking helps organizations identify and prevent potential security threats. (5)

A) Course Outcome: "CO5" / Cognitive Level: "Application"

B) Analyze how network scanning results help identify potential vulnerabilities and guide further security testing. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_3129:Digital Forensics

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_3129:Digital Forensics

#	Level	No of Questions	Marks (%)
1	Remember	1	5(10%)
2	Understand	3	15(30%)
3	Application	3	15(30%)
4	Analysis	3	15(30%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)